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Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Does the series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n+3}$ converge absolutely, converge conditionally, or diverge?

A series $\sum a_n$ converges absolutely (is absolutely convergent) if the corresponding series of absolute values, $\sum |a_n|$ converges. If the series converges, but the corresponding series of absolute values does not, then the series converges conditionally. Otherwise, the series diverges.

Find the terms of the corresponding series of absolute values.

$$\left| (-1)^n \frac{1}{n+3} \right| = \frac{1}{n+3}$$

The corresponding series of absolute values is similar to the harmonic series $\sum \frac{1}{n}$. The harmonic series diverges. Use the Limit Comparison

Test to determine the behavior of $\sum_{n=1}^{\infty} \frac{1}{n+3}$.

Suppose $a_n > 0$ and $b_n > 0$ for all $n \geq N$, for some integer N . The Limit Comparison Test states that if $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$

both converge or both diverge. Find the limit for $a_n = \frac{1}{n+3}$ and $b_n = \frac{1}{n}$.

$$\lim_{n \rightarrow \infty} \frac{n}{n+3} = 1$$

Therefore, since both series have positive terms, the series of absolute values has the same behavior as the harmonic series. The harmonic series diverges. Therefore, the series of absolute values diverges.

The Alternating Series Test states that the series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$ converges if the u_n 's are all positive, the positive u_n 's are eventually nonincreasing, and $u_n \rightarrow 0$.

Since n is always positive, then $u_n = \frac{1}{n+3}$ is positive for all values of n .

Find the ratio $\frac{u_n}{u_{n+1}}$. Substitute $n+1$ for n in the expression for u_n and divide.

$$\frac{u_n}{u_{n+1}} = \frac{n+4}{n+3}$$

If $u_n \geq u_{n+1}$, the ratio $\frac{u_n}{u_{n+1}}$ will be greater than or equal to 1 for all $n \geq N$ where N is some integer. The constant in the numerator is larger than the constant in the denominator. So the ratio will always be greater than 1, so $u_n \geq u_{n+1}$ for all values of n .

Find the limit of u_n as $n \rightarrow \infty$.

$$\lim_{n \rightarrow \infty} \frac{1}{n+3} = 0$$

All three conditions of the Alternating Series Test are satisfied. Therefore, the series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n+3}$ converges conditionally.